



ABSTRACT — MY PROJECT FEATURES

HANTER: Certificados al instante

The following abstract presents the formative project **HANTER: Certificados al instante**, developed under the technical competence guidelines of the ADSO program at SENA. It provides a clear, brief, and structured synthesis of the project's main elements.

1. Problem

Many organizations face challenges in managing employee certificates efficiently. Traditional manual processes for generating, tracking, and storing labor certifications are time-consuming, error-prone, and difficult to scale, leading to disorganization and reduced productivity within Human Resources departments.

2. Objective

HANTER is a web-based software system developed to automate and centralize the generation and management of employee certificates. The platform aims to provide a structured, secure, and role-based environment where HR personnel can create, retrieve, and manage certifications for each employee.

3. Methodology

The project was developed following an Agile approach, applying iterative phases from requirements analysis to implementation. The technology stack includes **Node.js** with **Express** for the backend REST API, **React** with **Vite** for the frontend, **Prisma ORM** with **SQLite** as the database engine, and **JSON Web Tokens (JWT)** for authentication and session management. Role-based access control restricts system functionality to authorized users such as Administrators and Supervisors.

4. Results

The system delivers a fully functional web application featuring secure login, protected routes, employee and role management modules, and a certificate generation workflow. The relational database model ensures data integrity across employees, users, and certificates. A formal software delivery and training session were carried out on May 12, 2026, at Sistemas Integrados SAS in Bucaramanga.

5. Conclusion

HANTER represents a significant step toward the digitalization of HR processes. By centralizing certificate management in a secure and accessible platform, the project reduces administrative overhead, minimizes human error, and improves operational efficiency within organizations. The successful delivery and adoption of the system demonstrate its practical value and readiness for real-world use.

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Firma del aprendiz
Quien presenta

Firma del instructor
Quien recibe